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Abstract

Electric leakage refers to an extremely important safety concern, which may lead to electric shocks, fire risks, and extensive damage to the electrical appliances and infrastructure. As the use of electrical systems in homes and industries increases, the issue of the safety mechanisms that are to be applied in this context to ensure safety has become more relevant than ever. This project is concerned with the design and development of the Electric Leak Detection System that is capable of detecting the leakage current and also provide real time alert to the users.

The proposed system makes use of sensors and a microcontroller to constantly measure the flow of electricity and identify any unusual variations that could be an indication of leakage. In cases where such conditions are detected, the system triggers alert facilities like a buzzer or LED to alert users instantly. The system is cost effective, efficient and easy to implement and is therefore applicable in practice.

Moreover, this project puts a strong emphasis on the combination of embedded systems and simple IoT principles to improve electrical safety. The general objective is to reduce risks, avoid accidents and offer a reliable solution to early detection of electrical fault. The results demonstrate that the system is capable of improving safety measures and can be further enhanced with advanced features such as remote monitoring and automated control.

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1. Introduction

1.1 Overview of IoT

The Internet of Things (IoT) is the concept of interconnecting ordinary objects and devices via the internet, so that they can collect, exchange, and process data. IoT is the combination of physical devices with the digital environment, where the former can communicate and interact with the latter without direct physical interaction. Such devices often include sensors, actuators and communication interfaces which gather real-time data about their surroundings and send it on to further processing or action.

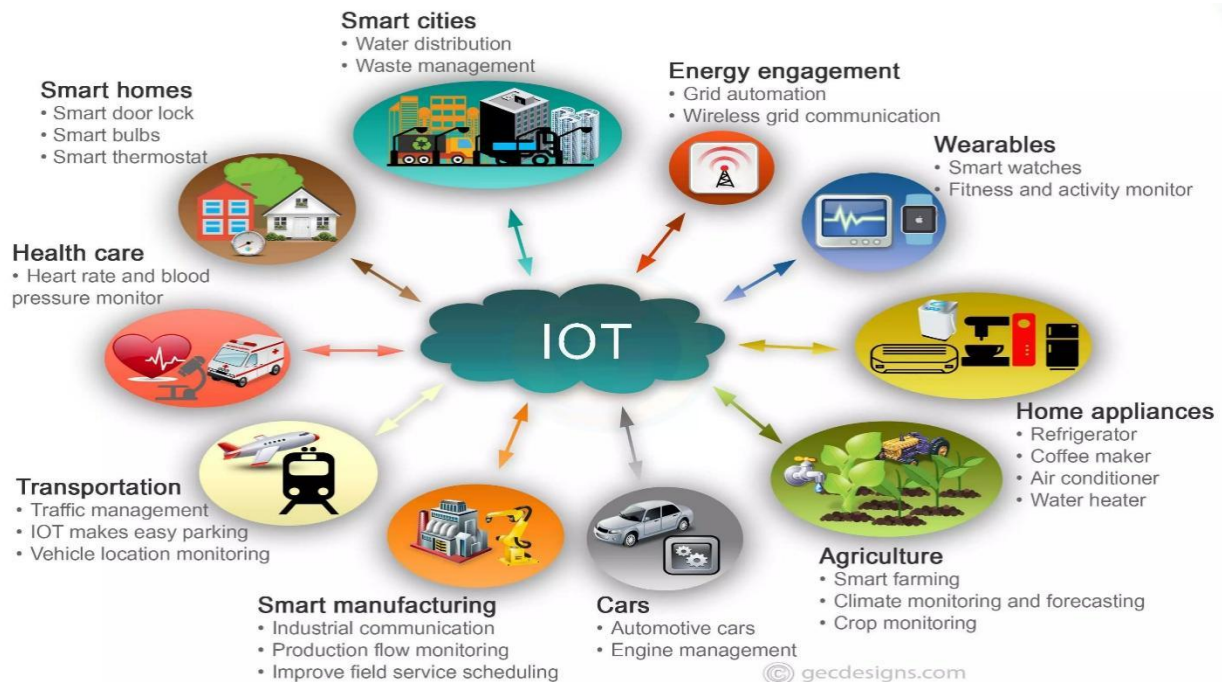


Figure 1 Types of IoT Devices

When it comes to industrial and domestic applications, IoT is essential in making things more efficient, automated, and safe. Remote monitoring and control of devices can be used to facilitate real-time decision-making and reduce the necessity of manual input. The advantages of IoT are enormous: savings on costs, convenience, and increased productivity to the development of completely new services and business models. Nonetheless, it also presents difficulties, including the security, privacy, as well as the difficulty of managing large-scale networks of connected devices. (Anon., u.d.)

1.2 Aim and Objectives

1.2.1 Aim

The main aim of this project is to design and develop an efficient and reliable Electric Leak Detection System that can identify leakage currents and provide real-time alerts to enhance electrical safety in residential and small-scale environments. (Anon., u.d.)

1.2.2 Objectives

To achieve the above aim, the following objectives are defined:

- To study and understand the concept of electric leakage and its associated risks
- To design a system capable of continuously monitoring electrical current flow
- To send both onsite and offsite alert(remote alert).
- To select and integrate appropriate hardware components such as sensors, microcontrollers, and alert devices. (Anon., 2020 jan 23)

1.3 Current Scenario

The entire system is intended to be cost effective, simple to install and applicable in small scaled applications, which can be easily accessed by ordinary users to apply during safety- related issues. In modern electrical systems, leakage monitoring is often performed by equipment like circuit breakers or special test equipment. Although these solutions are effective, they have a number of limitations:

- Expensive in the case of high-level monitoring.
- Poor accessibility by general users.
- Simple configurations do not have real-time monitoring.
- No proper remote monitoring system for small scale.

Leakage is a common problem that goes unnoticed in most households and small businesses until it results in equipment failure or even dangerous conditions. Also, current protective devices might not be able to offer a lot of information concerning present flow or the early warning. (Willam, 2022 nov 14)

Thus, there is a necessity of a cost-effective, real-time, and automated system that has the ability to detect leakage and alert in real-time to prevent potential damage.

1.4 Problem Statement

Electrical leakage is a serious problem that is sometimes hard to detect in its early stages with the use of traditional methods. The solutions available are either priced too highly, not flexible or they do not offer continuous monitoring capabilities to small scale applications. Consequently, users have a number of challenges:

- The risk of electrical hazards is increased.
- Inefficiency in energy consumption because of leakage that cannot be detected.
- Absence of automatic protection systems.
- Failure to detect early leakage.

Thus, there is a necessity of a cost-effective, real-time, and automated system that has the ability to detect leakage and respond in real-time to prevent potential damage and ensure safety. (Anon., u.d.)

1.5 Scope and Limitations

1.5.1 Scope

The current project is aimed at creating a prototype system that will be able to:

- Real time monitoring of current flow.
- Leakage detection, in terms of leakage threshold values.
- Remote Alert system.
- Delivering output via a serial monitoring interface.
- Proper workflow through esp32.

The system is mainly designed in the small scale, like in household circuits or laboratory systems, where cost and simplicity are a significant consideration. (Anon., u.d.)

1.5.2 Limitations

The system is rather functional, but it has some limitations:

- Poor accuracy because of sensors noise and calibration limitations.
- Not applicable to high voltage industrial systems that require other protection measures.
- Dependence of Wi-Fi connection .
- No long-term data storage through cloud or other service provider.

These limitations highlight areas for future improvement and system enhancement.

2. Background

2.1 System Overview

The electricity leak detector system is based on the principle of imbalance current detection. In normal conditions, the current flowing through the live wire should be equal to the return current through the neutral wire. If there is a difference between these two values that means it is leaking. (Anon., u.d.)

We measure the current flow with a current sensor and send it to a micro controller. The micro controller processes the signal and compares it to a threshold value. If leakage is detected:

- A buzzer is activated to provide an audible alert
- LEDs indicate the system status (normal or fault)

The system works in real-time and ensures immediate response, making it highly effective for preventing electrical hazards.

2.2 Tools and Technologies Used

The development of the Electric Leak Detection System involves a combination of hardware and software tools to ensure efficient design, implementation, and testing of the system.

2.2.1 Hardware Components

The following hardware components were used in this project:

2.2.1.1 Microcontroller (Arduino/ESP32):

Acts as the main processing unit of the system. It reads data from the sensor, processes it, and controls the output devices. (Andrei, u.d.)

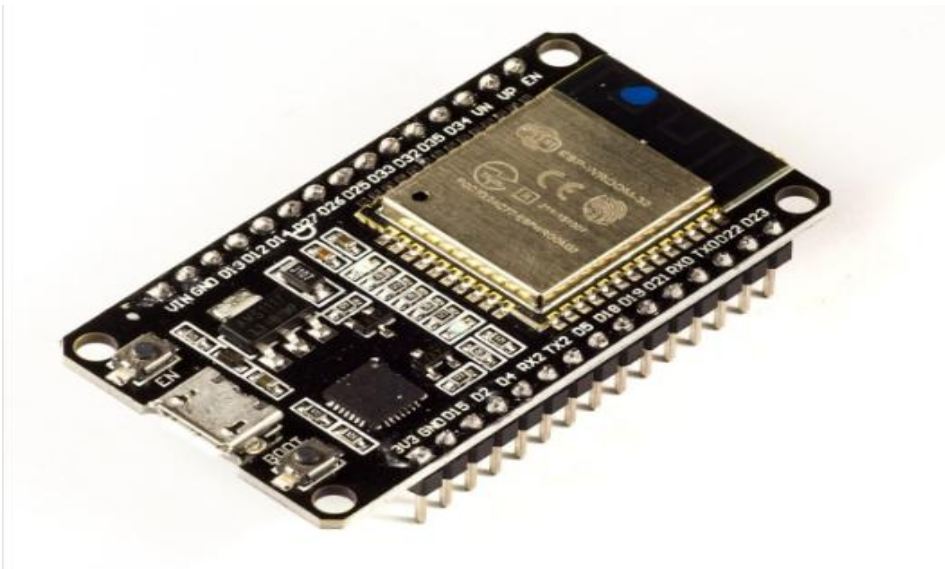


Figure 2 ESP 32 (Teel, n.d.)

2.2.1.2 Current/ Leakage Sensor ACS712:

Used to detect abnormal current flow or leakage in the electrical circuit. It continuously monitors the current and sends data to the microcontroller. (Raheel, u.d.)

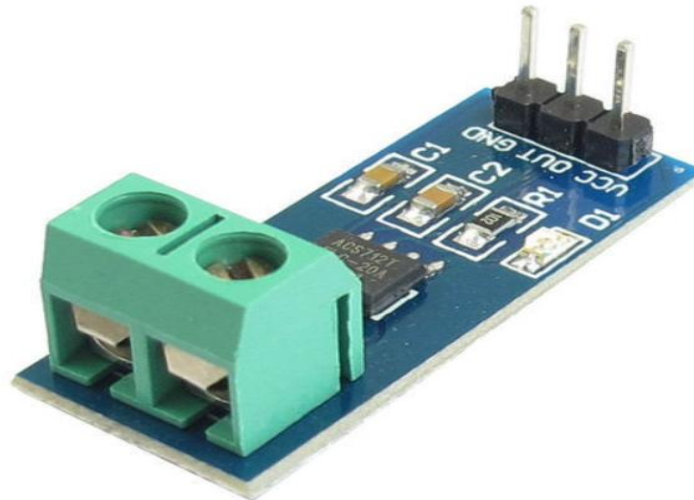


Figure 3 ACS712 (electric current sensor) (elprocus, n.d.)

2.2.1.3 Buzzer:

In the Electric Leak Detection System, the buzzer functions as an alert mechanism to notify users when electrical leakage is detected. When the sensor identifies abnormal current flow, the microcontroller sends a signal to activate the buzzer, producing a sound alarm. This immediate audible warning ensures that users are quickly informed about potential hazards, even if they are not directly observing the system.



Figure 4 buzzer

2.2.1.4 LED Indicator:

Provides a visual indication of system status (normal or leakage detected). (Anon., u.d.)

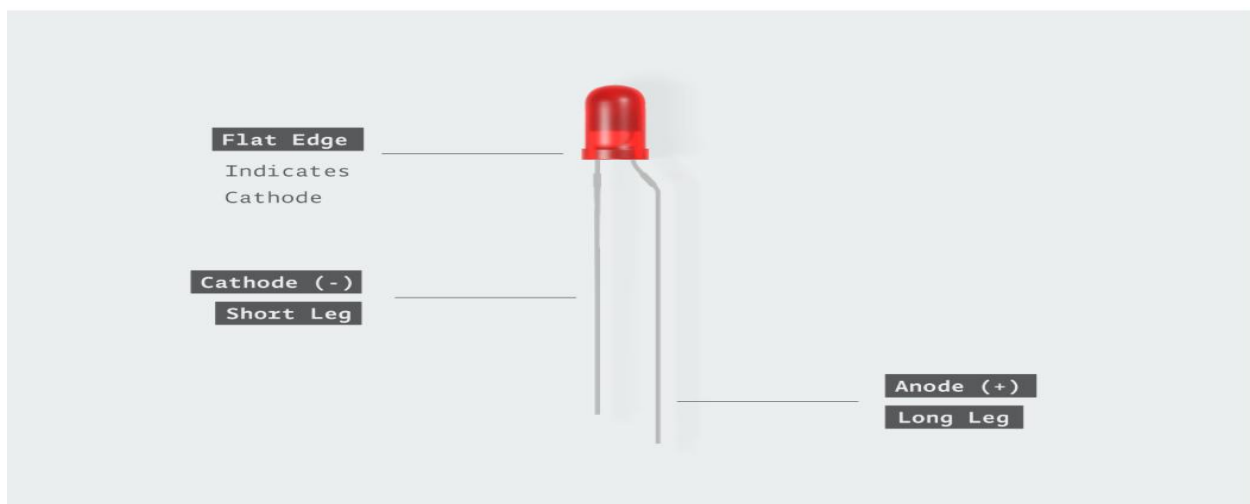


Figure 5 Led light (Arduino, 2022)

2.2.1.5 Power Supply:

Supplies the required electrical power to operate the system components.



Figure 6 Battery

2.2.1.6 Jumper Wires

Jumpers Small metal connectors that are used to open or close a part of a circuit. They have two or more connection points, which regulate an electrical circuit board. They are used to set the settings of computer peripherals, such as the motherboard. A jumper may be configured to have it on or off. (Wiltronics, 2022)

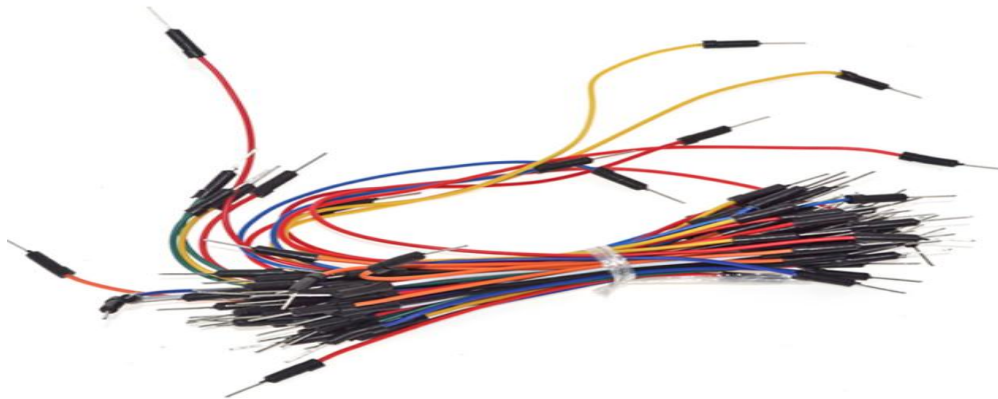


Figure 7 Jumper Wires (Wiltronics, 2022)

2.2.1.5 Breadboard

A breadboard is a reusable prototyping tool used to build and test electronic circuits without the need for soldering. It consists of a grid of interconnected holes where electronic components such as resistors, sensors, and microcontrollers can be easily inserted and connected using jumper wires. (ScienceDirect, 2022)

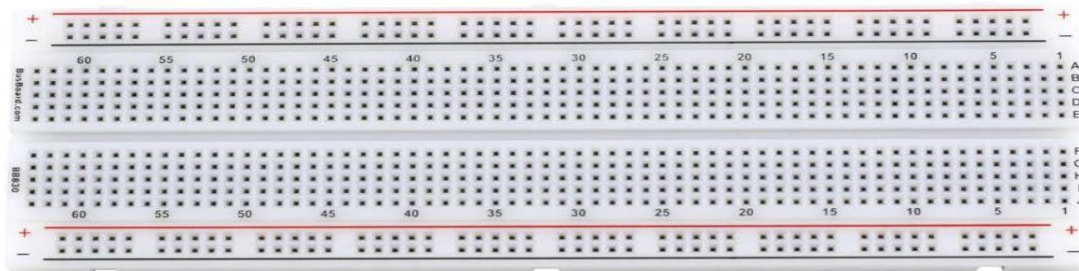


Figure 8: Breadboard

2.2.1.6 Live Wire

The live wire, also known as the phase wire, is the conductor that carries electrical current from the power source to the device. It is typically at a high voltage and is responsible for supplying energy to the circuit. In the context of this project, the live wire is the main line through which current flows, and any leakage from this wire is detected by the system. Proper handling of the live wire is essential for safety, as it can cause electric shock if touched. (Sparky, 2022)

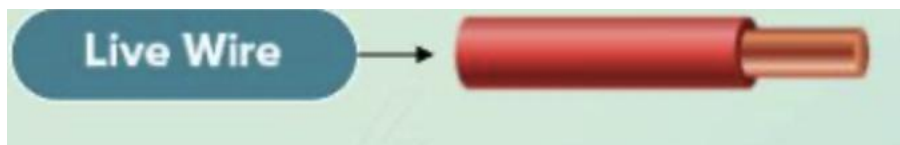


Figure 9: Red Live wire

2.2.1.7 Neutral Wire

The neutral wire is the conductor that carries current back from the device to the power source, completing the electrical circuit. It is usually at or near zero voltage compared to the ground. In a properly functioning system, the current flowing through the live wire should be equal to the current returning through the neutral wire. Any difference between these currents may indicate leakage. (geeksforgeeks, 2025)

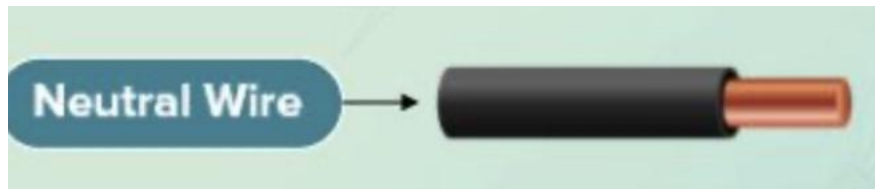


Figure 10: Neutral wire

2.2.1.8 BLYNK

Blynk is an IoT (Internet of Things) platform that still lets the user control and monitor devices such as ESP32, ESP8266, and Arduino Uno via a cell app over the web. It communicates with the hardware device via Wi-Fi and with the Blynk cloud server and app. The app allows users to build buttons, switches, gauges, and charts to remotely control LEDs, motors, fans and more. Ease of setup and the fact that you do not have to create a new mobile application makes Blynk a popular option for smart home systems, sensor monitoring, automation and for beginner IoT projects.

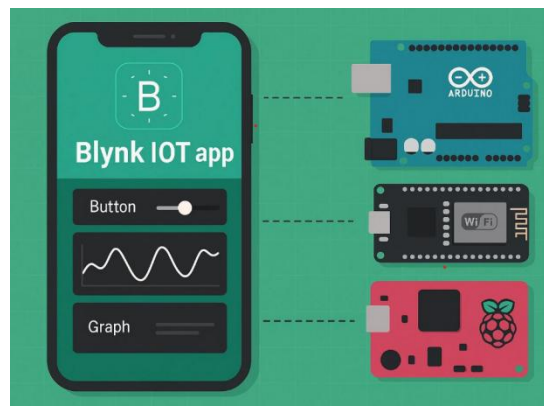


Fig 11: Blynk

2.2.2 Software Tools

2.2.2.1 Arduino IDE:

Used to write, compile, and upload code to the microcontroller. It provides a simple and user-friendly environment for programming. (Anon., u.d.)

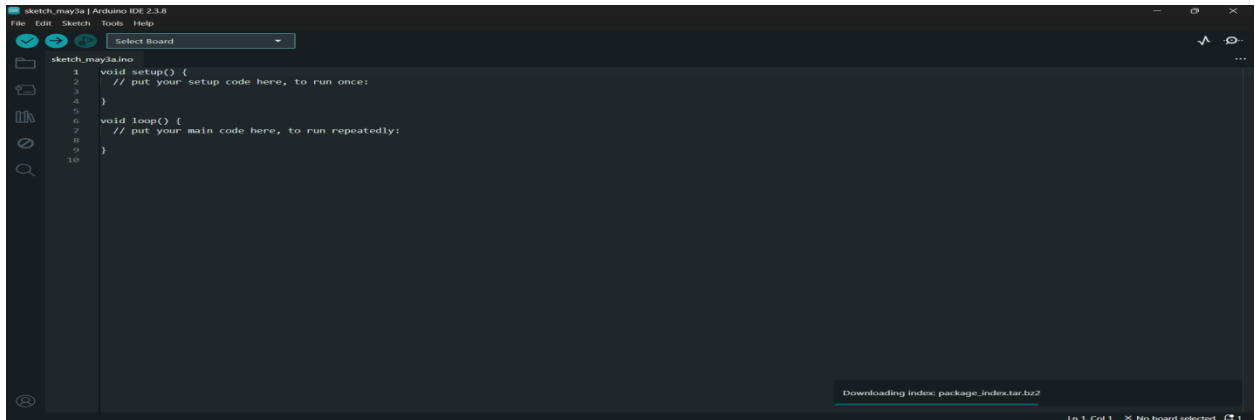


Figure 12: Arduino IDE

2.2.2.2 Embedded C / Arduino Programming Language:

Used for the development of the logic for detecting leakage and controlling the alert system.

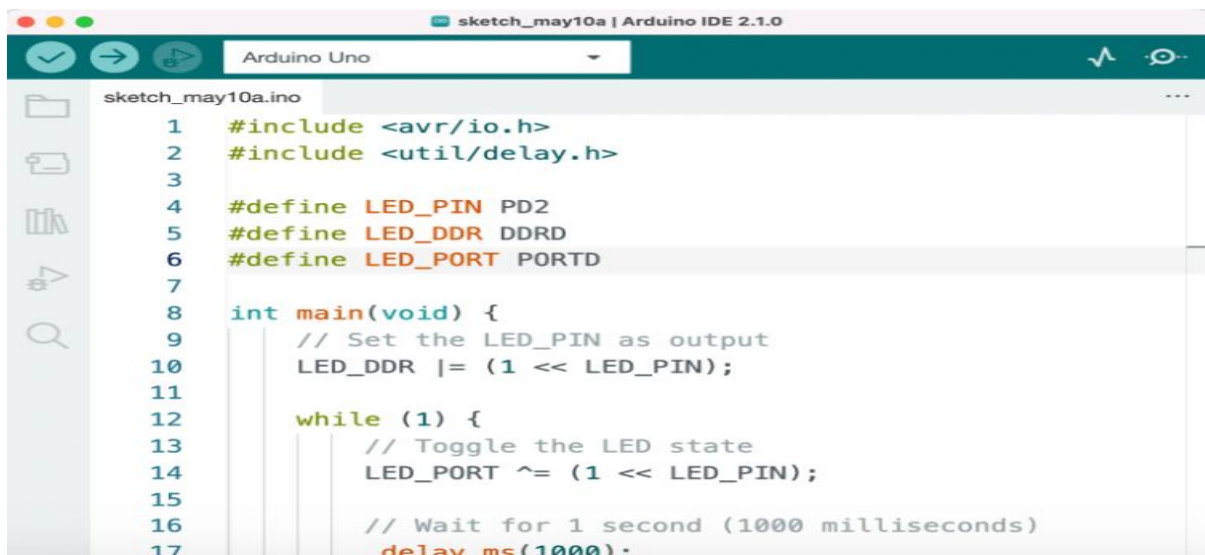


Figure 13: Arduino programming language (C) (Peter, 2023)

2.2.3 Draw.IO

Draw.io is proprietary software, created by Seibert Media, to make diagrams and charts. The program will allow you to use an automatic layout feature, or you can make your own layout. The drag-and-drop feature makes it simple to create a great looking diagram or chart. (Hope, 2025)

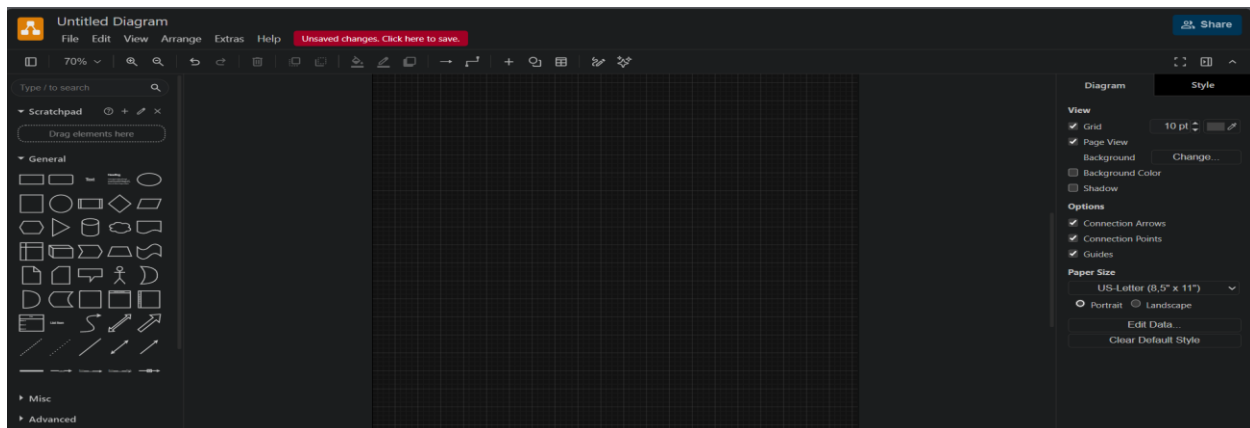


Figure 14 Draw.io Homepage

2.2.4 Fritzing

Fritzing An open-source hardware platform that aims to make electronics more accessible as a creativity medium. It offers software development tools, a community web site and similar services based on a platform such as Processing and Arduino. (fritzing, 2026)



Figure 15 Fritzing Website

2.3 System Architecture

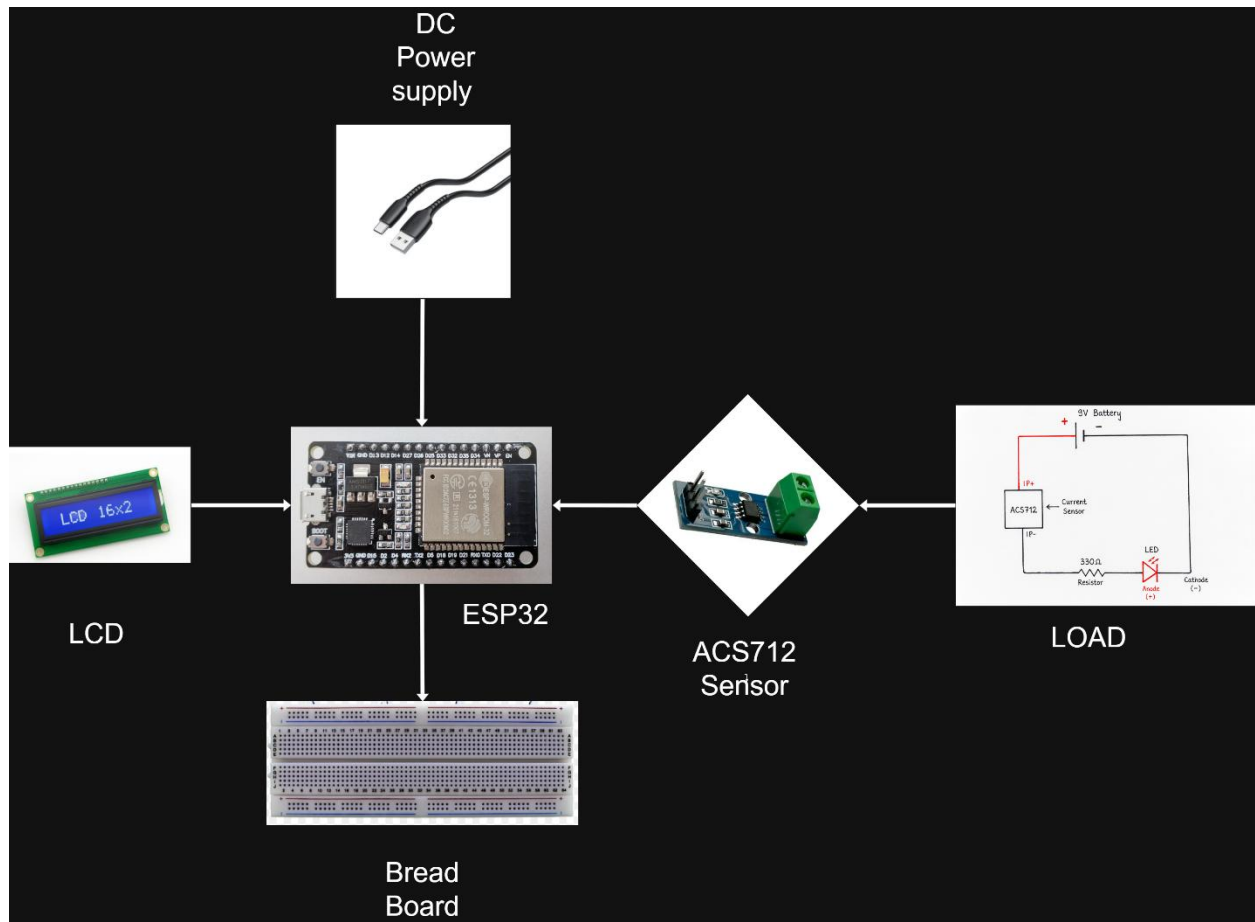


Figure 16 System Architecture of Electricity Detector

2.4 Block Diagram

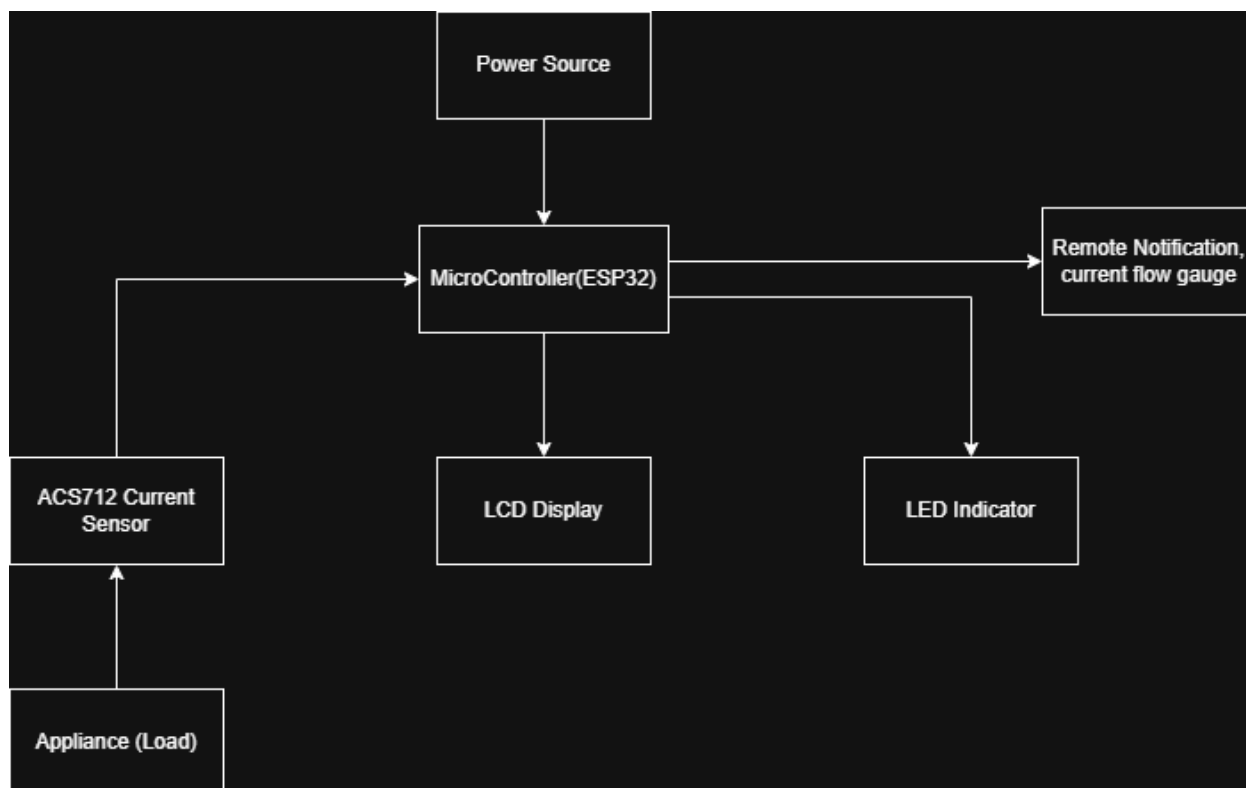


Figure 17 Block diagram of the Electricity Detector

2.5 Contribution of Group Members

Student's Name	Roles	Contribution
Himanshi Oli (Leader)	Proposal: Introduction, Conclusion System Development Report: Results and Findings Application Implementation: Designing	20%

	Presentation: Background	
Pranshu Shrestha	Proposal: Development Process System Development Report: Development, Further Work Application Implementation: Wiring, Coding Presentation: development	20%
Sanket Bhandari	Proposal: Background, System Design System Development Report: Background, Design Application Implementation: Coding Presentation:	

	Slides, Animation and Design	
Philip Kunwar	System Development Report: Introduction and References Application Implementation: Resource Management, Wiring	20%
Arun Kumar BC	System Development Report: Background, Appendices Application Implementation: Coding Presentation: Introduction and Conclusion	20%

Table 1: Contribution

3. Development

3.1 Project Overview

The high rate of electrical and electronic systems in residential, commercial and industrial contexts has augmented the relevance of ensuring electrical safety. Electrical leakage is one of the key issues in such systems and it can cause the wastage of energy, damage of equipment, and severe safety risks such as electric shocks and fire hazards.

In this project, the design and development of an electrical leakage detector and protection system with embedded technology is presented. The system will be based on ESP32 (the central processing unit), which will handle all the processing and ACS712 Current Sensor, which is used to measure the current passing through a circuit.

The system, aside from detecting, is able to monitor the current condition in real time via a mobile application using Blynk. It supports users to view data remotely, provides warning in abnormal conditions, and can monitor the system anywhere and anytime with the Internet, making the system smarter and more interactive.

3.2 Flowchart

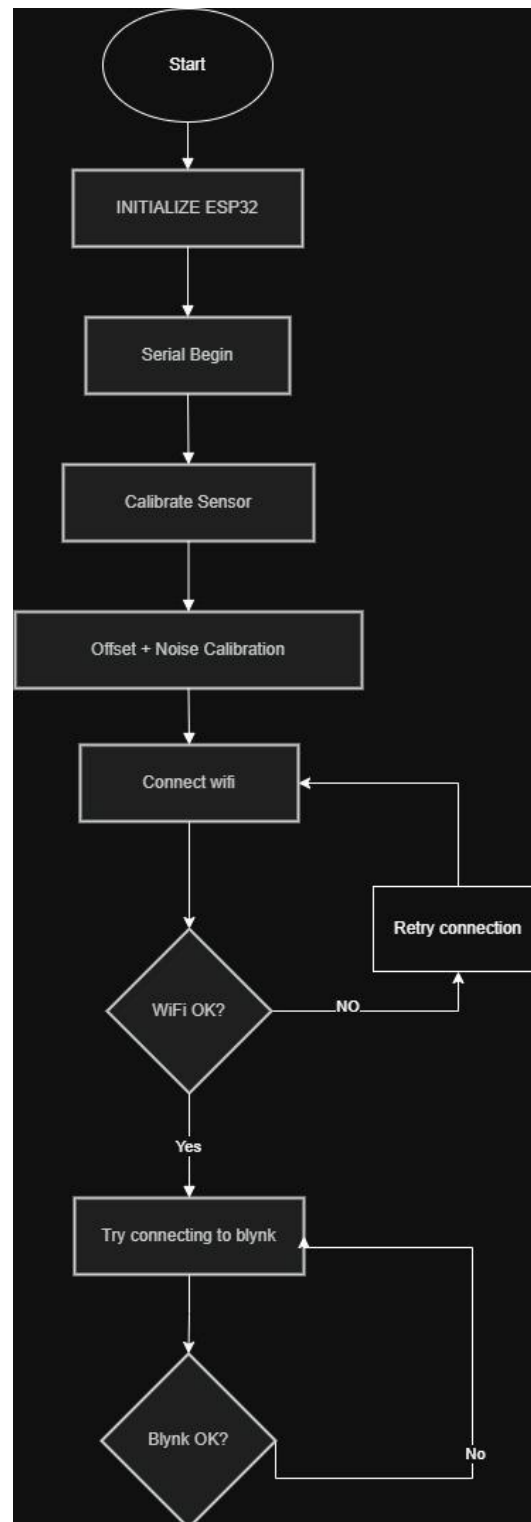


Figure 18: Flow chart of Electricity Detector Part 1

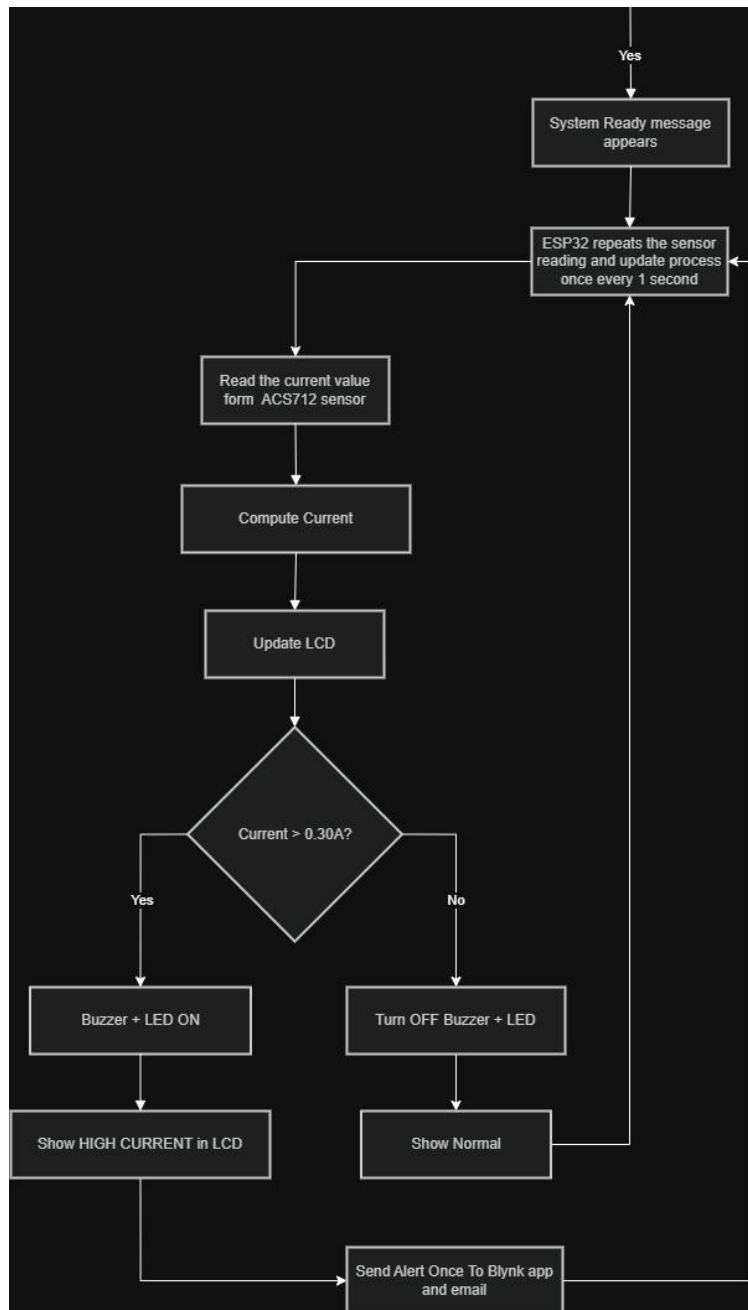
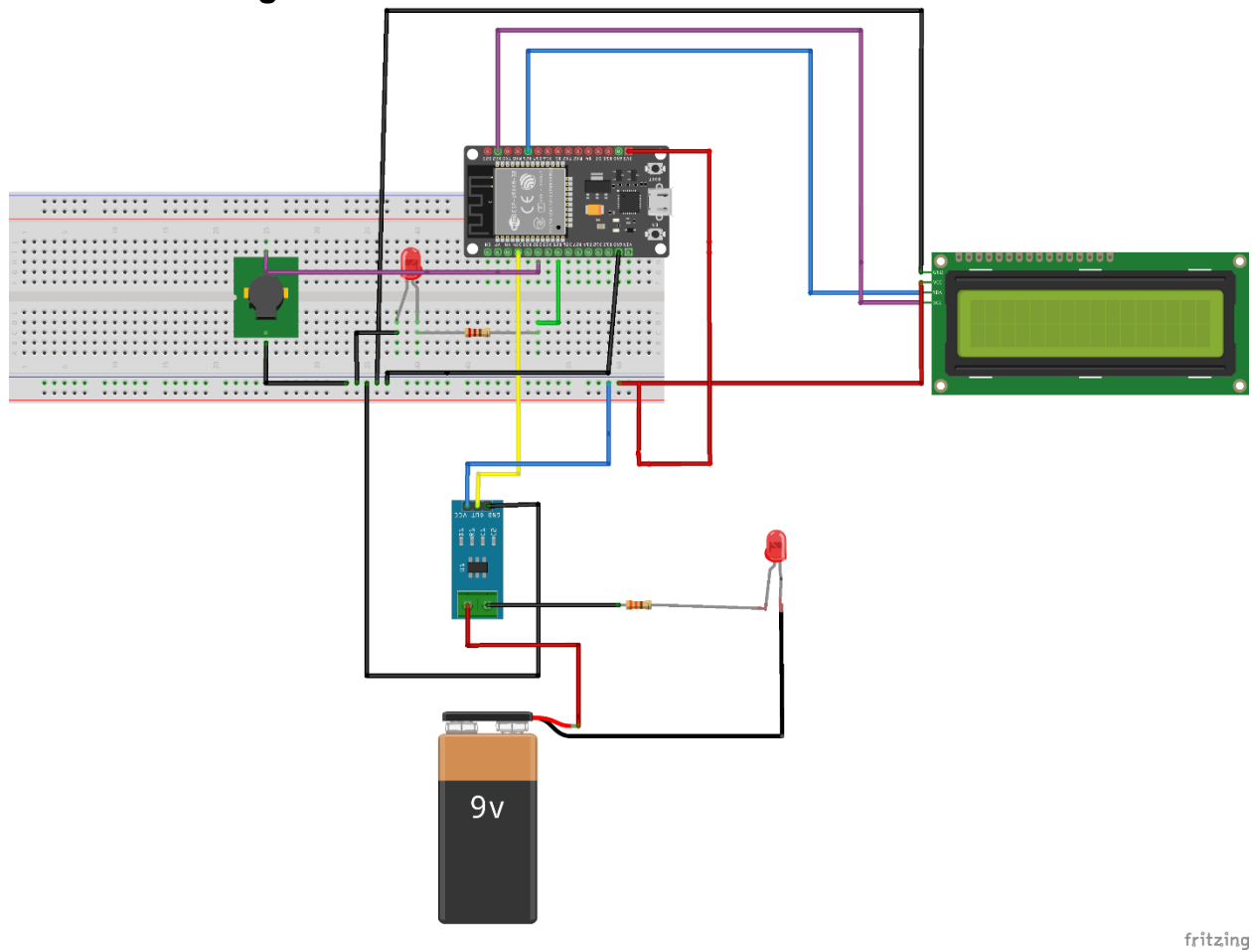


Figure 19: Figure 18: Flow chart of Electricity Detector Part 2

3.3 Circuit Design



fritzing

Figure 20 Circuit Diagram of Electricity Detector

3.4 Components and Their role in our project

3.4.1. ESP32:

Role: The ESP32 is the central microcontroller, which manages the whole system. It reads the ACS712 current sensor value and determines the functions of the components connected to it.

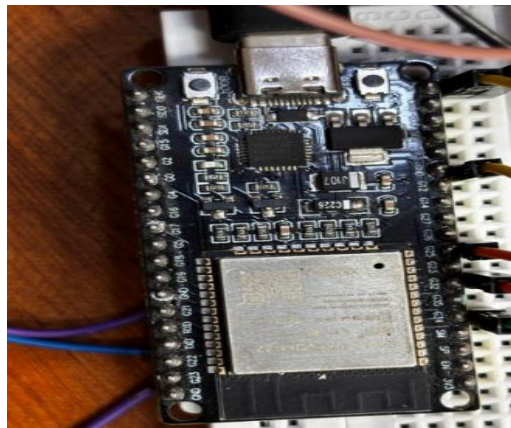


Figure 21 Esp32

It can also be in charge of the communication with the LCD and LED to be used as displays. (Anon., u.d.)

Connection: It is linked to the LCD and ACS712 sensor through its GPIO pins to process the inputs and outputs.

3.4.2. Blynk:

Role: The application Blynk lets you control and monitor devices via the Internet's connection between a mobile application and hardware devices such as ESP32 or Arduino Uno.

Connection: The hardware is connected to Wi-Fi, communicates with the Blynk cloud server and can then be controlled remotely using the Blynk app.

3.4.3. LED:

Role: LED is employed as a status indicator.

Connection: One of the GPIO pins of the ESP32 is connected to the LED via a current-limiting resistor.

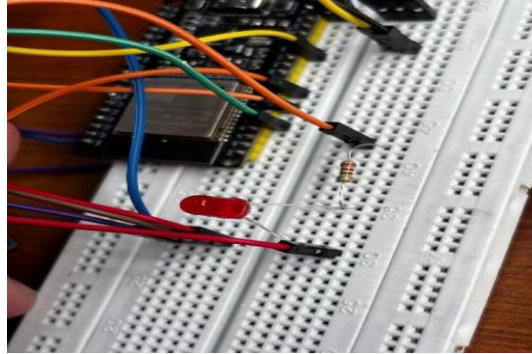


Figure 22 Led Light and Resistor

3.4.4. Resistor (for LED):

Role: The resistor is needed to ensure that the current flowing through the LED does not cause the LED to burn out due to excessive current.

Connection: The resistor is connected in series with the LED between the ESP32 GPIO pin and the anode (positive terminal) of the LED.

3.4.5. LCD:

Role: The LCD shows the current values of the ACS712 sensor and the state of the system (e.g., "Current OK," "Power Cut," and so on). It also gives the user real time feedback on the performance of the system.



Figure 23 LCD

Connection: The LCD is connected to the ESP32 through the I2C interface (or any other appropriate interface), usually using two pins: SDA and SCL as data and clock respectively.

3.4.6. ACS712 Current Sensor:

Role: ACS712 sensor measures the electric current passing in the circuit. It delivers an analogue voltage which is proportional to the current. The ESP32 will read this and when the current surpasses the set point, the ESP32 will cause the other components to function. (Anon., 2020 jan 23)

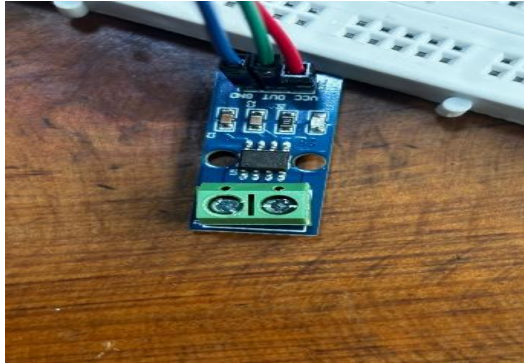


Figure 24 ACS712 Sensor

Connection: A pin of the ESP32 (e.g., GPIO34 or GPIO35) reads the current value, and it is connected to the ACS712 output pin (analogue voltage). The sensor ground and VCC pins are connected to the respective pins in the ESP32.

3.4.7. Load:

Role: Here the current flow is measured from battery, led, resistor which is our load that is connected to acs712 sensor.

Connection: The IP+ of acs712 sensor is connected to + of battery and IP- of acs712 sensor is joined to anode of led light through a resistor of 330ohm and also the cathode is joined to the -ve of the battery . This is a serial connection.

3.4.8. Buzzer

Role: Here a buzzer is used as an actuator that transmits sound by converting electrical energy sound energy .

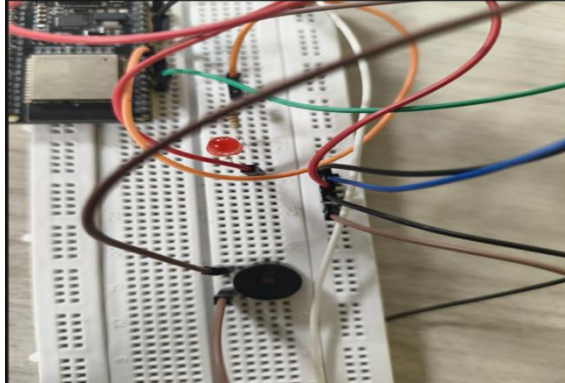


Figure 25 Buzzer in circuit

Connection: The buzzer is placed in the breadboard and the -ve side goes to common ground and the +ve side goes to the esp32 pin GPIO 32.

3.4.9. Jumper Wires:

Role: The necessary connections are needed to be made between components like the ESP32, LCD, ACS712 sensor, LED, buzzer in the breadboard.

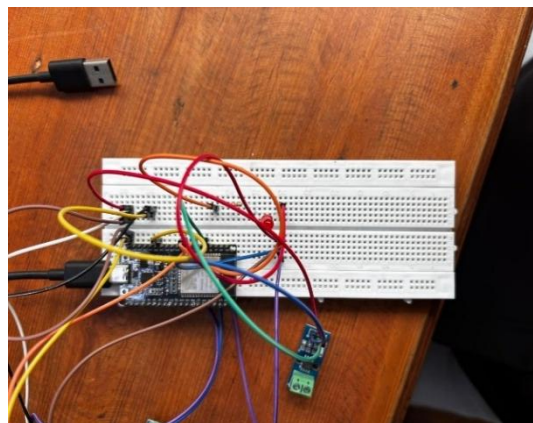


Figure 26 Jumper wires

Connection: Jumper wires are used to provide a connection between the components to each other in order to provide the required electrical pathways through which the signals would flow.

3.4.10. Breadboard:

Role: The components are mounted on the breadboard to create connections between them. It offers a fast and temporary means of connecting the circuit without soldering.

Connection: All the components (ESP32, buzzer, LCD, ACS712, and LED) are mounted and connected to the breadboard by using jumper wires. (Anon., u.d.)

3.5 Port to port component connection

3.5.1 ACS712 (Current Sensor)

Pin Connection on ESP32	Pin on Component	Function
5V	VCC	Powers the ACS712 sensor
GND	GND	Ground connection
GPIO34	OUT	Reads the latest measurement (analog input)

Table 2 : ACS712 port-port connection

3.5.2 Led

Pin Connection on ESP32	Pin on Component	Function
GPIO25	Anode	The LED is controlled by the value of the LED
Resistor (220Ω-330Ω)	Cathode	Limits the current to the LED

Table 3: Led port-port connection

3.5.3 LCD (I2C)

Pin Connection on ESP32	Pin on Component	Function
GPIO21	SDA	I2C data connection
GPIO22	SCL	I2C clock connection
5V	VCC	Power to the LCD
GND	GND	Ground connection

Table 4: LCD port-port connection

3.5.4 Buzzer

Pin Connection on ESP32	Pin on Component	Function
GPIO32	+ve side	Gives power to buzzer and also determines the function.
GND	-ve side	Ground connection.

Table 5 Buzzer Connection

3.5.5 Load

Components	Connection
9V Battery (+ve)	Ip+ of Current Sensor to '+ve' of 9volt battery
ACS712 Current Sensor (IP-)	Connected to 330 Ω Resistor
330 Ω Resistor	Connected to led Anode (+)
LED Cathode (-)	Connected to 9V Battery (-ve)

Table 6 Load Connection

4. Results and Findings

4.1 Overview

The project is a combination of ESP32 with other components such as the ACS712 Current Sensor, LED, LCD, and Blynk, which will be used to monitor the current flowing through the wire, control and observe the system based on threshold parameters. The system also aims to measure the current, display real-time data, and alert and send the current data to the Blynk app remotely in case of exceeding a set current limit, thereby protecting the safety of the devices connected to the system.

The system can read current from the ACS712 sensor, process the current values obtained by the ESP32, and transmit to Blynk, which can be used to view and monitor the data in real-time. The LED and LCD display offer feedback to the user locally about the current status of the system or its working condition, and Blynk allows remote access and notification to the status of the system via the internet.

4.2 Test Cases Table

Test 1: Showing the “High Current” in LCD

Objective	Confirm whether the system is able to identify high current and show it on the LCD.
Action	Simulate High Current by loading the circuit with more load.
Expected Result	The LCD shows the phrase High Current when the value of current surpasses the previously set value.
Actual Result	The LCD showed a high current as it should have done when the load was switched to a higher value and the current was more than the threshold.
Conclusion	Successful

Table 7: Testing 1



Figure 27: High Current

Test 2: Showing the “Normal” in LCD

Objective	Determine whether the system is able to detect status of current and to display it on the LCD.
Action	Did not pass any current and checked the lcd for normal Status.
Expected Result	LCD indicates its normal.
Actual Result	The LCD actually displayed the real-time current reading 'Normal'.
Conclusion	Successful

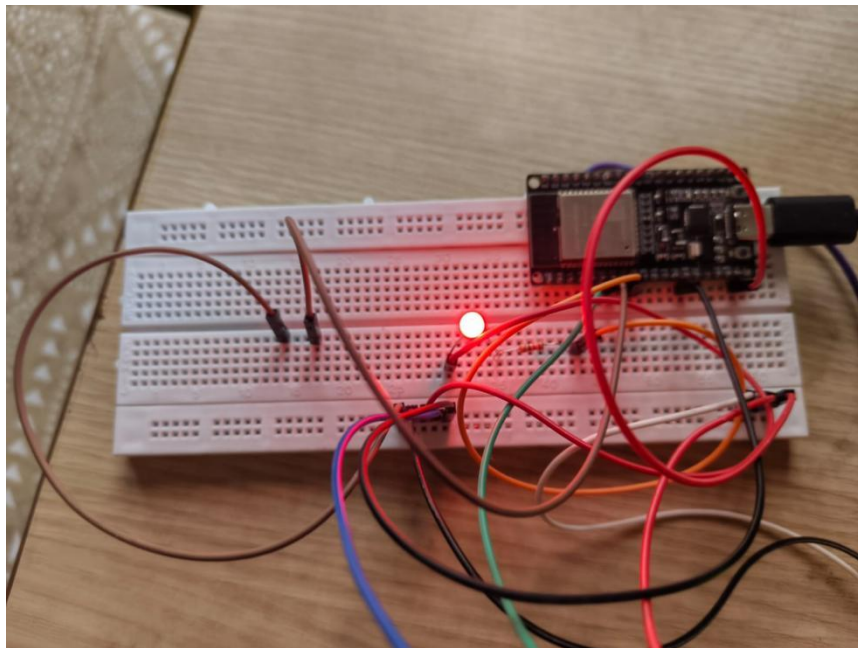
Table 8: Testing 2



Figure: Current flow is Normal

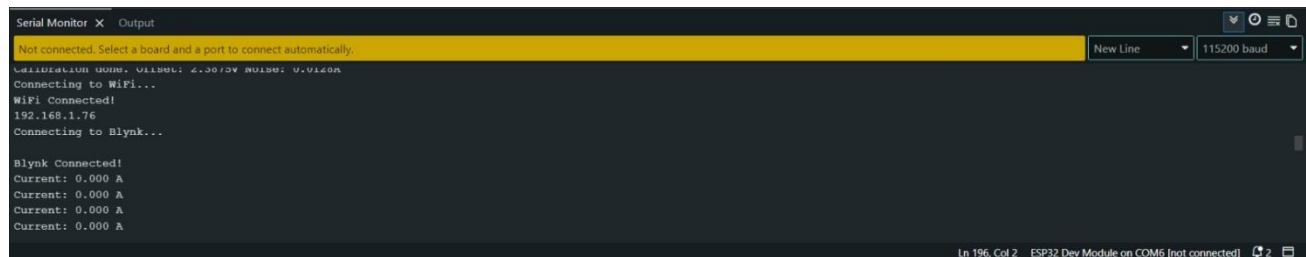
Test 3: LED bulb turns on high electricity flow

Objective	Check whether the LED works properly by Crossing the set threshold.
Action	Cross the threshold which we had applied and checked if the led is working or not.
Expected Result	When the current rises above the limit, the LED is expected to turn on.
Actual Result	As expected, the LED light was illuminated.
Conclusion	Successful

Table 9: Testing 3**Figure: LED turning On**

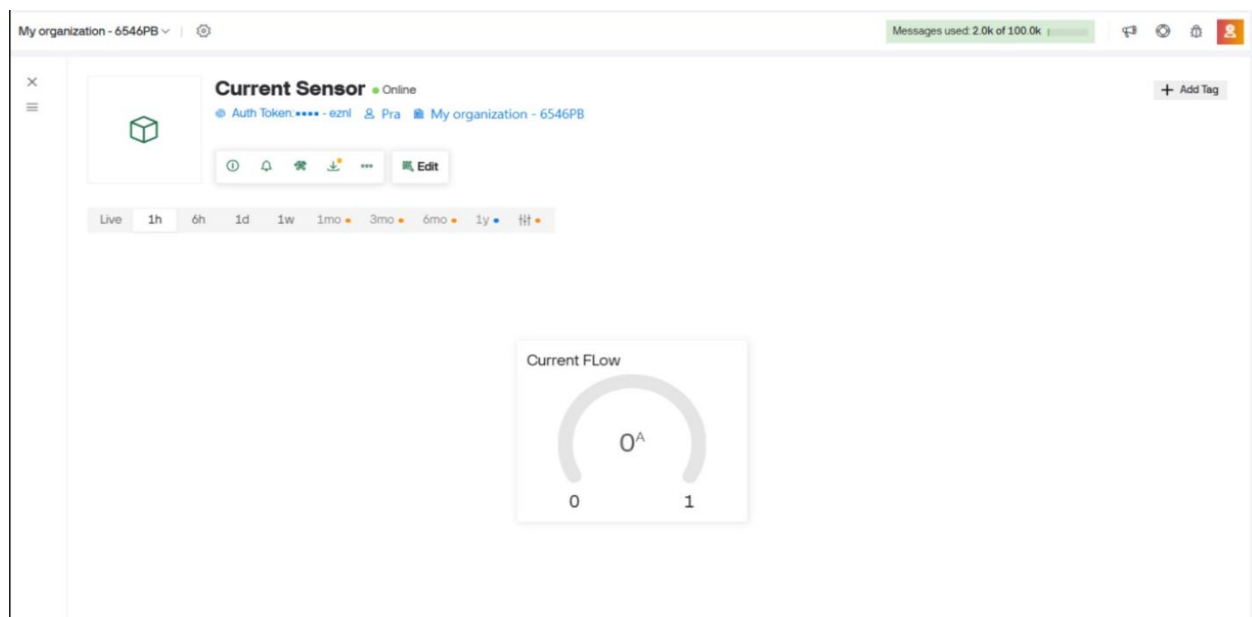
Test 4: Checking the serial monitor for Wi-Fi and Blynk connections

Objective	To check whether both Wi-Fi and Blynk are being connected through serial monitor
Action	We uploaded the code to show the status of Wi-Fi and also configured Blynk and ran the code to check their status.
Expected Result	Wi-Fi connecting and Wi-Fi- connected status first the Blynk connecting and Blynk connected status shown in the serial monitor
Actual Result	Serial monitor showed connecting Wi-Fi and Blynk and also was successfully connected and working.
Conclusion	Successful

Table 10 Test 4: Monitoring through Serial Monitor**Figure 28 Serial Monitor**

Test 5: Blynk App live Current flow

Objective	To make sure that the live current is shown simultaneously even in the Blynk app
Action	We encoded esp32 and also configured Blynk dashboard with a gauge with virtual pin 0 whose measurement unit is Amp.
Expected Result	The live current should be shown immediately without any delay.
Actual Result	Even though there was millisecond of delay it still worked.
Conclusion	Successful

Table 11 Test 5: Live Current Flow in Blynk App**Figure 29 Gauge when Current at rest**

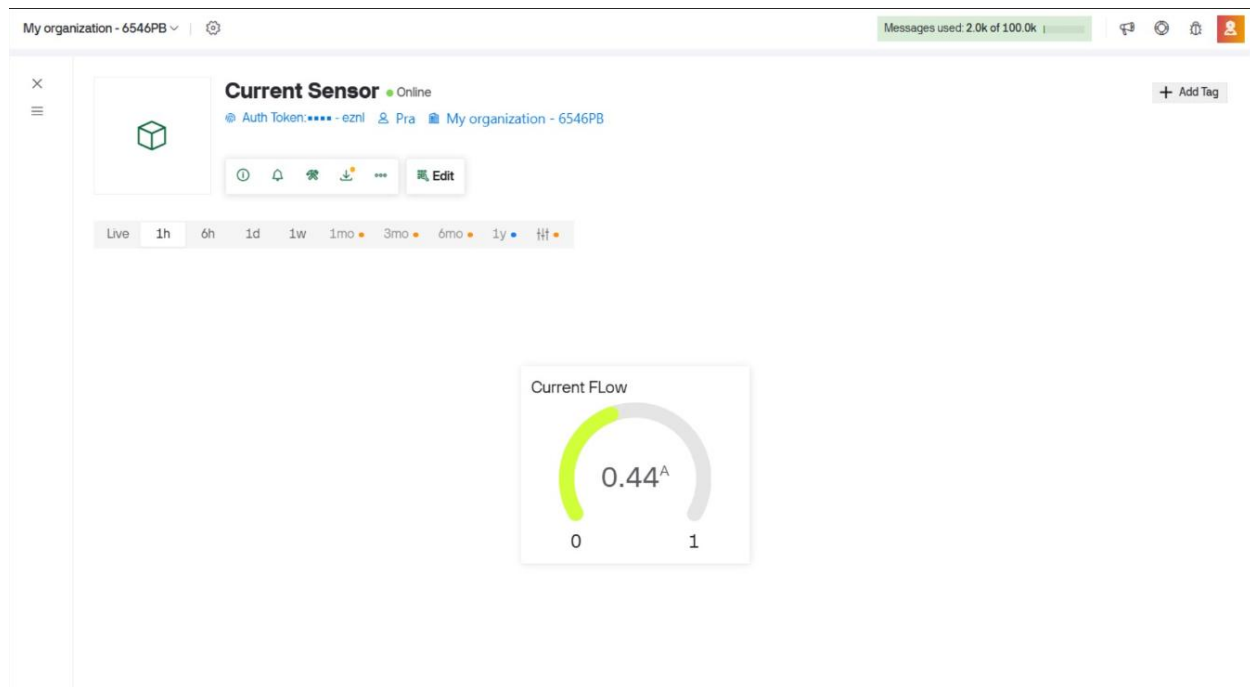


Figure 30 Gauge when current flow is measured

Test 6: Blynk App Alerting High Current

Objective	Check whether the system can correctly identify high current and show the status in Blynk mobile app.
Action	Increased the current flow beyond the set threshold value and checked the real-time data on the Blynk app connected to the ESP32.
Expected Result	If the current is over the limit the Blynk app should immediately display a high current warning or update current value to show abnormal conditions.
Actual Result	As expected, the Blynk app was able to show the current value in real-time, and that the current was past the safe level.
Conclusion	Successful

Table 12: Test 6: Checking app for Notification and alerts

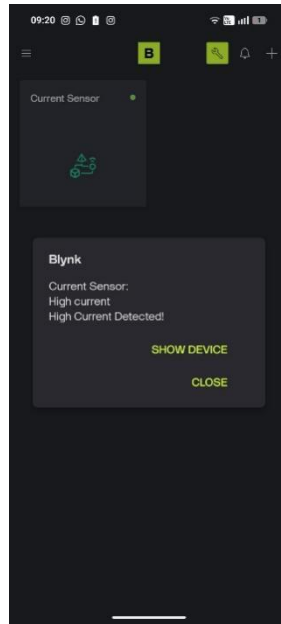
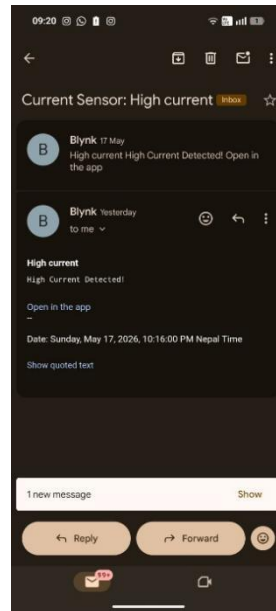


Figure 31: Alerting in Blynk App about High current

Test 7: Blynk App Alerting High Current in Email of user

Objective	Check whether the system is able to send an email notification when the current is above the threshold value using Blynk.
Action	Raised the current flow above the specified limit and observed the response from the system. Determined if the ESP32 sent a notification to Blynk and if the user received an email alert message.
Expected Result	If the current exceeds safe limit, the system should send an instant mail alert with Blynk with high current condition.
Actual Result	As expected, Blynk was able to send the user an email notification, informing that the value of the current has been exceeded.
Conclusion	Successful

Table 13: Test 7: Checking the email for Notification**Figure 32: Blynk App Alerting High Current in Email of user**

5. Further Work

While the IoT current monitoring and alert system have been successfully developed and tested, there are some enhancements that could be included in future for better system performance, reliability and functionality.

5.1 Mobile Application Enhancement

Currently system is remote monitoring/notification system via Blynk. With mobile application development in future versions, a custom dashboard featuring real-time current readings, alert history and device status can be created. Enhanced security and user-friendliness would result from further features like user authentication, device control, and graphical data visualization.

5.2 Data Logging and Cloud Storage:

Currently the system only provides real-time monitoring and alerts. One can extend the work in the future by incorporating cloud-based platforms like Google Firebase or Thing Speak to store sensor information on a continuous basis. This would provide the ability

to analyse the data over the long term, generate historical reports, and be able to monitor electrical device performance over time.

5.3 Battery Backup System

The system is currently powered only by a continuous electricity supply. A rechargeable battery back-up or UPS module would allow the system to continue operating during a power outage. This enhancement will improve reliability particularly in critical environments where continuous current monitoring is essential for safety and protection.

5.4 Multiple Sensor Integration

The project involves only one ACS712 Current Sensor Module to currently monitor one electrical line. There is room for expansion to multiple current sensors input to allow monitoring of multiple devices or circuits. This will make the system more suitable for industrial applications, smart homes or laboratories where multiple electrical loads must be supervised simultaneously.

5.5 AI-Based Fault Detection

The system can be upgraded in the future with Artificial Intelligence or a machine learning algorithm to automatically analyse electricity consumption. The system could determine if there was a fault or electrical problem before it happened by studying the abnormality in the current flow. Such a predictive monitoring technique would enhance the safety of the system, lower maintenance expenses and boost operational efficiency.

5.6 Compact PCB Design

The current prototype is connected with breadboard connections, which are primarily used for testing and demonstration. Future work may include the design of an original Printed Circuit Board (PCB) to construct a compact, robust and polished system. A PCB based design would help minimize loose connections, ensure circuit stability and make the project deployable to the real world and viable for commercial use.

6. Conclusion

To conclude, the current monitoring and alert system for the IoT industry had been designed and implemented with the help of an ESP32 microcontroller, ACS712 current sensor, LCD display, buzzer, LED indicators, and Blynk IoT platform. The system can measure real-time current, and can display measured value on LCD directly, transmit the measured value to the Blynk app remotely and transmit alert at the time of abnormal condition.

The project illustrates the benefits of using IoT technology to enhance electrical monitoring and access to live data, both locally and remotely. Immediate alerts were given on site with the LED and buzzer, and improved user awareness and convenience with the mobile notification system. The ability to integrate hardware and cloud-based monitoring also contributed to the automation and ease of use of the system.

Overall, the project achieved its main objectives of creating a low-cost, efficient, and user-friendly current monitoring solution. It also offered hands-on experience and practical knowledge of embedded systems, interfacing with sensors, communication over the Internet of Things (IoT), and real-time monitoring technologies. While the system has some drawbacks, it is a good base for future enhancements, including advanced data logging features, improved security, automatic protection features, and mobile app development.

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8. Appendix

8.1 Code

```
#define BLYNK_TEMPLATE_ID "TMPL67-jMz3Zy"  
#define BLYNK_TEMPLATE_NAME "Current Sensor"  
#define BLYNK_AUTH_TOKEN "A5L2FHUa718nc_RPWMm8Rb0yMEfWeznl"
```

```
#include <WiFi.h>  
#include <BlynkSimpleEsp32.h>  
#include <Wire.h>  
#include <LiquidCrystal_I2C.h>
```

// WiFi

```
char ssid[] = "prasanna02_2.4";  
char pass[] = "CLFE2CF417";
```

// LCD

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

// Pins

```
const int sensorPin = 34;  
const int buzzerPin = 32;  
const int ledPin = 25;
```

// ACS712

```
const float sensitivity = 0.185;  
const float refVoltage = 3.3;  
const int adcMax = 4095;
```

// Calibration

```
float offset = 0;
```

```
float zeroNoise = 0;
```

// Threshold

```
const float leakLimit = 0.30;
```

// Alert flag

```
bool notified = false;
```

// Timer

```
BlynkTimer timer;
```

// CALIBRATION

```
void calibrateSensor() {
```

```
    lcd.clear();
```

```
    lcd.print("Calibrating...");
```

```
    Serial.println("Calibrating sensor...");
```

```
    float sumVoltage = 0;
```

```
    for (int i = 0; i < 300; i++) {
```

```
        int raw = analogRead(sensorPin);
```

```
        sumVoltage += (raw * refVoltage) / adcMax;
```

```
        delay(2);
```

```
    }
```

```
    offset = sumVoltage / 300.0;
```

```
float sumCurrent = 0;
for (int i = 0; i < 300; i++) {
    int raw = analogRead(sensorPin);
    float v = (raw * refVoltage) / adcMax;
    sumCurrent += abs((v - offset) / sensitivity);
    delay(2);
}
zeroNoise = sumCurrent / 300.0;
```

```
Serial.printf("Calibration done. Offset: %.4fV Noise: %.4fA\n", offset, zeroNoise);
```

```
lcd.clear();
lcd.print("Cal Done");
delay(1000);
lcd.clear();
}
```

// SENSOR FUNCTION

```
void readAndReport() {

    float total = 0;
    for (int i = 0; i < 100; i++) {
        int raw = analogRead(sensorPin);
        float v = (raw * refVoltage) / adcMax;
        total += (v - offset) / sensitivity;
    }

    float current = abs(total / 100.0) - zeroNoise;
```

```
if (current < 0) current = 0;
```

```
Serial.printf("Current: %.3f A\n", current);
```

// LCD LINE 1

```
lcd.setCursor(0, 0);
```

```
lcd.print("Current:   ");
```

```
lcd.setCursor(0, 0);
```

```
lcd.print("Current:");
```

```
lcd.print(current, 2);
```

```
lcd.print("A");
```

// Blynk

```
Blynk.virtualWrite(V0, current);
```

// ALERT

```
if (current > leakLimit) {
```

```
    digitalWrite(buzzerPin, HIGH);
```

```
    digitalWrite(ledPin, HIGH);
```

```
    lcd.setCursor(0, 1);
```

```
    lcd.print("HIGH CURRENT  ");
```

```
    if (!notified) {
```

```
        Blynk.logEvent("high_current", "High Current Detected!");
```

```
        notified = true;
```

```
    }
```

```
} else {
```

```
digitalWrite(buzzerPin, LOW);
digitalWrite(ledPin, LOW);

lcd.setCursor(0, 1);
lcd.print("Normal    ");

notified = false;
}
}

// SETUP
void setup() {
  Serial.begin(115200);

  pinMode(buzzerPin, OUTPUT);
  pinMode(ledPin, OUTPUT);

  digitalWrite(buzzerPin, LOW);
  digitalWrite(ledPin, LOW);

  lcd.init();
  lcd.backlight();

  calibrateSensor();

  // WIFI CONNECTION
  Serial.print("Connecting to WiFi");
  lcd.clear();
```

```
lcd.setCursor(0, 0);  
lcd.print("WiFi Connecting");  
  
WiFi.begin(ssid, pass);  
  
while (WiFi.status() != WL_CONNECTED) {  
    delay(500);  
    Serial.print(".");  
    lcd.setCursor(0, 1);  
    lcd.print(".");  
}  
  
Serial.println("\nWiFi Connected!");  
Serial.println(WiFi.localIP());  
  
lcd.clear();  
lcd.setCursor(0, 0);  
lcd.print("WiFi Connected");  
lcd.setCursor(0, 1);  
lcd.print(WiFi.localIP());  
  
delay(1500);  
  
// BLYNK CONNECTION  
Serial.println("Connecting to Blynk...");  
lcd.clear();  
lcd.setCursor(0, 0);  
lcd.print("Blynk Connecting");
```

```
Blynk.config(BLYNK_AUTH_TOKEN);
Blynk.connect();

while (!Blynk.connected()) {
  delay(500);
  Serial.print(".");
  lcd.setCursor(0, 1);
  lcd.print(".");
}

Serial.println("\nBlynk Connected!");

lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Blynk Connected");
lcd.setCursor(0, 1);
lcd.print("System Ready");

delay(1500);
lcd.clear();

// Timer
timer.setInterval(1000L, readAndReport);
}

// LOOP
void loop() {
```

```
Blynk.run();  
timer.run();  
}
```